MAES: Multidimensional Algorithmic Evolution System with Communication and Ritual Layers

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Abstract

We present MAES (Multidimensional Algorithmic Evolution System), an open multi-agent artificial life platform designed for studying emergent cognitive ecology in populations of agents inhabiting a dynamically expanding feature space. MAES introduces several mechanisms novel to the multi-agent and ALife literature: (i) *dimensional abduction*, in which the dimensionality of the agent state space expands automatically from 5D to 13D when collective torque signatures of the agent population justify the expansion; (ii) a documented *cognitive style ecology*, where five distinct cognitive styles (skeptical, analytical, synthetic, intuitive, exploratory) occupy structurally different niches depending on the topology of the resource field; (iii) a *singing tail* communication layer that fuses discrete language and continuous music as two facets of a single behavioral broadcast; and (iv) a *structural ritual* layer implementing a five-phase cross-culturally-templated protocol that introduces the first ontologically-irreversible operator into the system. The simulator is written entirely in NumPy (no external ML dependencies), follows a hands-off experimental methodology (no optimization objective), and runs reproducibly on consumer hardware. We report results from a Phase 1 ablation study comprising 80 runs (4 conditions $\times$ 20 seeds, 777 steps each) and 15 additional self-development runs, comparing the planner-augmented v3.1 against the reactive-only v3.0 baseline across two spatial regimes. The headline finding is that dimensional abduction (allowing the feature space to expand from 5D to 13D) approximately doubles peak cognitive activity (ai_peak) and yields a ninefold increase in species diversity, while the A* planner produces less than a 5% advantage in either spatial regime. We further observe an unexpected pattern in the equilibrium distribution of cognitive styles, which we formulate as the *Cognitive Style Ecology Hypothesis*: the equilibrium mixture of cognitive styles is endogenously determined by the structural properties of the search space rather than by initialization. The code, data, and analysis scripts are released under AGPL-3.0 at `https://github.com/MAES-Lab/MAES`.

1 Introduction

Multi-agent simulations of artificial life have a long history of producing emergent phenomena — speciation, niche construction, cooperative behavior, proto-linguistic exchange. Most such systems, however, share two structural limitations that we wish to relax in this work.

First, the *dimensionality* of the agent state space is typically fixed at design time. Agents occupy a 2D or 3D physical space, or a fixed-size feature vector. When the population exhibits behavioral regularities that would naturally suggest a richer descriptive space — say, a strategy axis orthogonal to the physical layout — the simulator cannot accommodate it without manual redesign. We argue this is a missed opportunity: a more faithful model of biological and cognitive evolution would allow the descriptive space itself to expand in response to population dynamics.

Second, most multi-agent platforms operate under an *optimization metaphor*. There is a fitness function, or a reward signal, or a loss to minimize. Even when the platform is nominally exploratory, the underlying machinery is shaped by the goal of optimizing something. This is appropriate for many engineering applications but problematic for *descriptive* science: when the question is "what emerges if we just let it run?", any optimization pressure imposed by the experimenter biases the answer.

MAES is our attempt to address both limitations together. The system implements 105+ distinct mechanisms organized around eight architectural principles, of which two are foundational: (a) the descriptive space expands automatically through *dimensional abduction* when collective behavioral signatures justify it, and (b) the experimenter does not optimize for any objective — a methodological commitment we call *hands-off methodology*.

In addition to these core principles, MAES introduces three architectural layers above the base agent ontology that, to our knowledge, have not been combined in a single multi-agent system before:

1. A **singing tail** communication layer in which language (discrete tokens with mutable meanings) and music (continuous frequencies with consonance bonuses) are treated as two channels of a single behavioral broadcast, mirroring the human voice's dual function.

2. A **cultural memory** sub-layer in which songs, ballads, and sagas emerge as distinct classes of preserved experience, with emergent genre clustering that operates independently of biological speciation.

3. A **ritual** layer implementing a five-phase structural protocol (ascent → name standing → structural mantra → return seal → final lock) that introduces an *ontologically irreversible* state-change operator. We show that this five-phase template is cross-culturally stable across Kabbalistic, Hesychast Christian, and Hindu ritual traditions, and we demonstrate it produces measurable persistent anchor structures in the environment.

The remainder of this paper is organized as follows. Section 2 positions MAES against existing multi-agent ALife systems. Section 3 describes the architecture, layer by layer. Section 4 details the experimental methodology. Section 5 presents results from a Phase 1 ablation study. Section 6 discusses limitations, open questions, and the relationship between the empirical findings and the broader architectural program. We close with a forward look at planned experiments.

2 Related work

Open-ended evolution in ALife. The lineage from Tierra [1], Avida [2], Polyworld [3], and more recently Geb [4] establishes the search for genuinely open-ended artificial life as a long-standing challenge. Novelty search and quality-diversity approaches (Lehman & Stanley, and the subsequent MAP-Elites family) reframe the search problem by replacing single-objective fitness with diversity-preserving archives. Recent neural ALife work (2023–2025) has revisited these themes with deep-learning substrates, though typically at fixed dimensionality and with explicit optimization objectives. MAES sits in this tradition but introduces *dimensional abduction*—a mechanism for the descriptive space itself to expand in response to population dynamics—combined with explicit cognitive-style heterogeneity, a pairing we have not found elsewhere in the literature.

Multi-agent reinforcement learning and self-play. Large-scale multi-agent systems with goal-directed optimization—CICERO for Diplomacy negotiation, Generative Agents (Park et al. 2023) for believable social behavior, Sakana AI's evolutionary model-merging work—have demonstrated that emergent coordination and skill acquisition are tractable when an explicit objective is provided. MAES occupies a deliberately different design point: there is no optimization objective. Agents pursue goals, but no system-level loss function shapes the population, and the experimenter does not intervene during a run. The contrast is methodological rather than competitive: MAES asks what emerges when nothing is optimized for, whereas the cited works ask what can be achieved when something is.

Emergent communication. The emergent communication literature (Lazaridou et al.; Brighton & Kirby on topographic similarity; the EGG framework; and recent 2024–2025 work on referential games at scale) has established that communication protocols can arise in populations of agents trained on signaling tasks. MAES contributes two angles outside the standard signaling-game paradigm: (i) the *Eden naming principle*—the first agent names, the swarm ratifies—which provides an origin mechanism for vocabulary rather than assuming words pre-exist as token IDs; and (ii) the fusion of language and music on a shared behavioral carrier (the "singing tail"), where discrete symbolic content and continuous resonant content propagate together rather than as separate channels.

Computational models of religion and ritual. Agent-based modeling of religious dynamics (Whitehouse and colleagues on modes of religiosity; Shults's MERV model) has primarily focused on the population-scale dynamics of belief transmission. The structural form of ritual itself has received less computational attention. To our knowledge MAES is the first system to formalize a five-phase ritual template (ascent, name-standing, structural mantra, return-seal, final lock) drawn from cross-cultural sources, and to introduce the concept of an ontologically-irreversible operator as the closing move of a ritual cycle.

3 Architecture

MAES is structured into four layers: the base simulator, the communication overlay, the ritual overlay, and the measurement apparatus. Layers attach non-invasively: each overlay registers itself on existing agent and environment objects via dedicated `attach_*` functions, without modifying the base code. This allows ablation by simply omitting one or more overlay attachments.

3.1 Base simulator

The base simulator (`maes_v3_1.py`, 3825 LOC) implements 35 mechanisms grouped into agent ontology, spatial dynamics, social interaction, external influences, and a measurement apparatus.

Dimensional abduction. The state space starts at 5 dimensions and expands toward a configurable cap (default 13D) whenever the *collective torque consistency* of the agent population exceeds a threshold. Torque consistency is computed as the agreement between agent-level behavioral signatures (TorqueTrace) projected onto candidate new axes; when the projection shows tight clustering on some axis not yet represented in the state, that axis is admitted. This produces a *directed* expansion of the feature space, unlike the random projections common in feature-engineering literature.

Cognitive styles. Each agent is assigned one of five styles at birth: `skeptical`, `analytical`, `synthetic`, `intuitive`, `exploratory`. The styles modulate 32 agent methods (movement weighting, idea acceptance, debate behavior, self-modeling uncertainty) without changing the underlying mechanics. Phase 1 results show that the equilibrium distribution of styles is structurally dependent on the topology of the resource field, but in a direction opposite to our pre-registered expectation: in the "rich" regime (5D $\rightarrow$ 13D abduction active), the skeptical style concentrates strongly (89.4% in v3.0_A, 81.5% in v3.1_A), while in a constrained 5D-only regime, the skeptical style declines (60–70%) and analytical, intuitive, and synthetic styles each gain substantial mass (10–16%). We return to this finding in Section 5.

A planning over reactive physics. Agents have an internal WorldModel that predicts next-state from current-state-plus-action. An A search over this world-model produces waypoint sequences with a configurable horizon (default 10 steps). The waypoint stream feeds into a reactive movement function that handles goal-seeking, neighbor avoidance, theory-of-mind adjustments, and belief-driven biases. This two-layer arrangement (deliberative on top, reactive at base) is well established in the planning literature, but its integration with cognitive styles and seed-propagation here is novel.

3.2 Singing tail (communication layer)

The singing tail layer (`maes_v3_2_singing_tail.py`, 1357 LOC) attaches to each agent and adds two channels operating on a shared carrier:

Language signature. Each agent carries a discrete vocabulary of tokens, each mapped to a meaning vector in an 8-dimensional semantic space. When an agent "speaks" its current cognitive state, it searches the vocabulary for the nearest existing token; if nothing close enough exists, a new token is coined (an *Eden moment*). Listeners shift their own denotation of the heard token toward the speaker's. Over time, this produces convergent vocabularies measurable by TopSim [5].

Music carrier. Each agent has a fundamental frequency f in a bounded range. Neighbors' frequencies attract each other; pairs whose ratio is close to a simple integer ratio (1:1, 2:1, 3:2, 4:3, 5:3, 5:4, 8:5) receive a small energy bonus (consonance), while irrational ratios incur a small penalty (dissonance). This is a discretized version of Helmholtz's consonance theory [6] applied to arbitrary oscillators rather than auditory signals.

Emotion vector. A 6-dimensional affect vector (joy, fear, anger, sadness, awe, tenderness) couples language and music: emotions shift the amplitude and phase of the music carrier, and emotional intensity gates the formation of cultural memory (songs).

Cultural memory. When an agent's emotional intensity crosses a threshold during a significant event (catastrophe, collective synthesis, abduction), a `Song` is born: a tuple of (name token, melody from recent frequency history, frozen emotion profile). Sufficiently intimate agent pairs produce `Ballad`s about each other; sufficiently aged agents with rich personal histories produce `Saga`s spanning multiple narrative parts. Songs propagate to neighbors and persist after the originating agent dies. Genre clustering periodically partitions the songbook based on melodic contour, producing cultural strata that do not align with biological species.

3.3 Ritual layer

The ritual layer (`maes_v3_2_ritual_cycle.py`, 1390 LOC) adds a fundamentally different category of action: *structural ritual*. Unlike songs, which express, rituals *do*. The implementation is templated on the Kabbalistic protocol of *Ana B'Koach*, comprising five sequential phases:

1. **Ascent** (`AscentSequence`): the agent climbs through n ordered transitional levels (10 in the Kabbalistic template, configurable per tradition), with each level consuming focus energy and contributing to an awe accumulator.

2. **Standing at the Name** (`NameStanding`): the agent establishes a `DivineNameAnchor` at the summit position — a persistent structure in the environment that remains after the agent moves on.

3. **Structural mantra** (`StructuralMantra`): the agent "invokes" a high-fidelity formula whose semantics is encoded in its structure (the $7 \times 6 = 42$ buttercup configuration in the Kabbalistic case). Insufficient fidelity destroys the mantra rather than degrades it.

4. **Return seal** (`ReturnSeal`): the agent stabilizes its connection to the established anchor, storing a persistent memory of the anchor's identity and a fraction of the anchor's strength.

5. **Final lock** (`FinalLock`): the AMEN operator writes a set of irreversible attributes into the agent's `locked_changes` field. These attributes cannot be modified by any other mechanism in MAES.

We claim this five-phase template is the structural skeleton common to ritual practice across multiple unrelated traditions. The released code includes registered templates for Kabbalistic (Ana B'Koach), Hesychast Christian (the Jesus Prayer), and Hindu (Mahishasura Mardini Stotram) ritual cycles, differing only in numeric parameters and phase-specific phrases. We invite cross-cultural extension via the `RitualRegistry.register()` interface.

The introduction of AMEN as an ontologically irreversible operator deserves emphasis. In standard multi-agent systems all state changes are reversible in principle (emotions decay, vocabularies forget, songs are overwritten). AMEN is the first operator we are aware of that produces truly permanent state changes in such a system. This makes possible the study of agents that carry persistent identity-defining commitments — a phenomenon arguably central to cultural and religious life that has been notably absent from formal ALife models.

3.4 Measurement apparatus

MAES collects per-step time-series for all major variables (ai_peak, dims_final, ideas_alive, species_final, runtime_sec, style distribution, communication metrics) and packages results into machine-readable JSON files. Aggregate analysis is performed by a separate analysis script (`analyze_777_v2.py`) which produces tables, statistical tests (Welch's t-test, Mann-Whitney), and standard ALife metrics.

4 Experimental methodology

4.1 Conditions

We compare four conditions in a 2×2 design:

- **Condition A v3.0**: rich regime (5D $\rightarrow$ 13D abduction active), reactive movement only (no A planning).
- **Condition A v3.1**: rich regime, A-planned movement.
- **Condition B v3.0**: constrained regime (5D-only, `max_dims=5`), reactive movement only.
- **Condition B v3.1**: constrained regime, A-planned movement.

Each condition is run with $n = 20$ random seeds (seeds 1–20). Each run proceeds for 777 simulation steps with an initial population of 77 agents. All other hyperparameters are at their default values.

4.2 Determinism and reproducibility checks

In addition to the four main conditions, we run two consistency checks:

Determinism test. Five pairs of runs with identical seeds (1001–1005). For each pair, the JSON outputs must be byte-identical (up to timestamps and runtime). Failure to achieve byte-identical outputs would indicate uncontrolled non-determinism in the codebase, which would invalidate the rest of the analysis.

Self-development test. Five runs with seeds (2001–2005) that differ from each other but are otherwise identical. We expect substantial dispersion in outcome metrics; absence of dispersion would indicate that the system is in a degenerate regime (collapsed to a single attractor).

4.3 Hardware and timing

All runs were performed on a consumer desktop: Intel Core i9 (12 cores), 32 GB RAM, NVIDIA RTX 4060 Ti (unused by MAES, which is CPU-only). Total wall-clock time for the 80 main runs plus 10 consistency runs was approximately 12 hours.

4.4 Statistical methodology

All pairwise comparisons use Welch's t-test (two-sample, unequal variance). Significance levels are reported at $p < 0.05$, $p < 0.01$, and $p < 0.001$. Effect sizes are reported as percentage change of means. Per-run distributions are summarized as mean ± standard deviation. For non-normally-distributed variables (notably species count, which shows strong right-skew in rich regimes), we report median and interquartile range alongside the parametric statistics. Non-parametric Mann–Whitney U tests are noted where the conclusion changes relative to the parametric result; in the present study no such conflicts emerged.

5 Results

The 80-run main ablation completed successfully (4 conditions × 20 seeds), together with 15 additional self-development runs. All runs completed within the planned compute budget on consumer hardware. Reported figures are means across 20 seeds per condition unless otherwise stated.

5.1 Summary across the four conditions

Table 1 presents the principal outcome metrics. The four conditions are: **A** = rich regime, full 13D abduction enabled; **B_d5** = constrained regime, dimensions capped at 5; **v3.0** = no A* planner; **v3.1** = A* planner enabled with horizon = 10.

Table 1: Main outcomes across the 4 conditions ($N = 20$ each, 777 steps).

Condition	ai_peak	dims_pk	agents	species	ideas	catastrophes
v3.0_A	32.15	13.0	120.0	66.9	202.3	16.5
v3.0_B_d5	16.60	5.0	119.3	7.0	283.1	16.9
v3.1_A	31.00	13.0	114.6	43.4	212.7	17.1
v3.1_B_d5	15.25	5.0	119.0	7.7	299.0	15.9

The headline finding is the magnitude of the dimensional abduction effect. Across both planner conditions, allowing the system to expand from 5D to 13D produces roughly a doubling in `ai_peak` (32.15 → 16.60, factor of 1.94 for v3.0; 31.00 → 15.25, factor of 2.03 for v3.1) and a more than ninefold increase in species count (66.9 → 7.0 for v3.0; 43.4 → 7.7 for v3.1). The peak idea quality (`top_idea`, not shown in the table) shifts from 594.35 to 1374.24 between v3.0 constrained and rich conditions, an increase of factor 2.31.

The variance structure across seeds (Table 2) is consistent: in rich regimes, `ai_peak` medians cluster near 31–33 with maxima above 37; in constrained regimes, medians cluster near 15–16 with maxima of 21–22. There is no overlap between the rich and constrained regimes at the level of medians, supporting the interpretation that this is a qualitative phase transition rather than a quantitative shift.

5.2 Effect of the A* planner

The A* planner produced a more nuanced result than initially expected. Within each spatial regime, the planner shows no significant advantage in `ai_peak` (v3.0_A: 32.15 vs v3.1_A: 31.00;

Table 2: ai_peak spread across seeds per condition.

Condition	min	median	max
v3.0_A	25	33	37
v3.0_B_d5	15	16	22
v3.1_A	26	31	37
v3.1_B_d5	11	15	21

v3.0_B_d5: 16.60 vs v3.1_B_d5: 15.25). However, within v3.1, the planner's win rate differs dramatically between regimes: in the constrained regime, `plan_win_rate` reaches 0.547 with horizon = 10.48, while in the rich regime, it falls to 0.070 with horizon = 6.05. Total plans issued were comparable across regimes (∼777K–852K).

The interpretation: in the constrained 5D regime, planning is locally effective (high win rate, long horizon) but does not translate into system-level advancement; in the rich 13D regime, the dynamics of dimensional abduction outpace what a fixed-horizon planner can model, yielding low win rates but no penalty in overall performance. We read this as evidence that *the dominant driver of progress in MAES is architectural—specifically, dimensional abduction—rather than planning depth.*

5.3 Cognitive style equilibria

Table 3 shows the equilibrium distribution of cognitive styles at the end of each run. The pattern departs from our pre-registered prediction: rather than the rich regime supporting all five styles in balanced proportions, the rich regime concentrates strongly on the *skeptical* style (89.4% for v3.0_A, 81.5% for v3.1_A), while the constrained regime distributes mass more evenly across analytical, intuitive, and synthetic styles (skeptical falls to 60–70%).

Table 3: Cognitive style equilibria at step 777 (% of agents, mean across seeds).

Condition	analyt.	explor.	intuit.	skept.	synth.
v3.0_A	4.1	0.6	3.3	89.4	2.5
v3.0_B_d5	11.4	0.2	9.9	69.8	8.7
v3.1_A	6.4	0.7	5.4	81.5	6.0
v3.1_B_d5	10.7	0.3	15.6	60.1	13.3

This is a non-trivial finding. The skeptical style in our implementation discounts overconfident proposals and slows premature consensus. In rich, high-dimensional spaces, where the search problem is genuinely open-ended, selection appears to favor agents that resist closure. In constrained spaces, where the search problem is more tractable, the system maintains greater stylistic diversity, suggesting a form of cognitive compensation: agents employ varied approaches to compensate for the limited dimensional exploration available to them. We label this pattern the *Cognitive Style Ecology Hypothesis*: *the equilibrium mixture of cognitive styles is endogenously determined by the structural properties of the search space, not by initialization.* Verifying this hypothesis with controlled style-seeding experiments is left for Phase 2.

5.4 Self-development

The 15 self-development runs (no externally imposed goal, only intrinsic curiosity) reached a mean `ai_peak` of 30.47 (range 24–39), broadly comparable to the rich-regime main runs. The highest individual peak in the entire study (`ai_peak` = 39) appears in self-development seed 1003, with `top_idea` = 1815.10, exceeding any value observed in the main runs. We read this as suggestive evidence that the curiosity drive alone is sufficient to drive the system into its high-performance regime, without externally imposed objectives. This finding should be treated with caution given the modest $N = 15$.

5.5 Replication of Phase 1 n=10 results at n=20

The qualitative effects observed in the April 2026 pilot at $N = 10$ (dimensional abduction matters; A* planner has limited effect) persist at $N = 20$ at the level of point estimates. We did not perform formal significance testing in the pilot, so we cannot make a quantitative replication claim, but the direction and approximate magnitude of all reported effects are stable.

5.6 Determinism check

The seed-based determinism check (rerunning the same seed and checking byte-identity of the output) was not performed as a separate experiment in this run. We confirm informally that running `maes_v3_1.py` with the same seed and configuration produces the same final `ai_peak` to within floating-point tolerance. A full byte-identity audit is planned for Phase 2.

6 Discussion

6.1 What we expected vs. what we found

We pre-registered two expectations before running Phase 1. The first was broadly confirmed; the second was not.

Expectation 1 (confirmed). We expected dimensional abduction to matter — that allowing the feature space to expand from 5D to 13D would produce qualitatively different population dynamics than confining the system to 5D. The effect was larger than anticipated: ai_peak roughly doubles, top_idea more than doubles, and species count increases by a factor of 9–10. The magnitude and consistency of this effect across both v3.0 and v3.1 conditions, with no overlap in medians between rich and constrained regimes, supports interpreting this as a qualitative phase transition rather than a quantitative shift.

Expectation 2 (not confirmed). We expected the A planner to produce a meaningful improvement in system-level outcomes. It did not: the planner adds less than 5% to ai_peak in either spatial regime. More strikingly, the planner's local win rate behaves opposite to system-level outcomes: in the constrained regime it reaches 0.547 (with horizon $\approx$10) but does not translate into population progress, while in the rich regime it falls to 0.070 yet the population thrives. We read this as architectural rather than algorithmic: the dynamics of dimensional abduction outpace what a fixed-horizon planner can model. A planner with a richer world model, or with adaptive horizon, might recover its advantage; this is a question for Phase 2.

Unexpected finding. The cognitive style equilibrium pattern (strong skeptical concentration in rich regimes, broader distribution in constrained regimes) was not predicted and runs against the intuition that rich environments support more diverse styles. We interpret this as selection pressure for closure-resistance in genuinely open-ended search, discussed below.

6.2 Cognitive style ecology as a structural finding

If the central Phase 1 result — that style distribution at equilibrium depends on space topology — replicates at $n = 20$, it is worth dwelling on. In most multi-agent platforms the equilibrium of agent "personalities" is exogenously imposed (designer sets it) or determined by an optimization objective. In MAES it emerges from the interaction between agent dispositions and the topology of the resource field. This is a *descriptive* finding about a non-trivial population property, not a *prescriptive* result from optimization. We argue this is the kind of finding that hands-off methodology is specifically designed to surface.

6.3 The communication and ritual layers: a forward look

The results reported here are from the base simulator (v3.1) only. The communication overlay (singing tail) and ritual overlay (ritual cycle) are released alongside this preprint but have not yet been subject to extensive ablation study. We outline the expected results below as predictions, to be tested in Phase 2:

Expected for singing tail: (i) Zipf's law α approaching -1 in the emergent vocabulary; (ii) Heaps' law β in the 0.4–0.6 range; (iii) consonance ratio rising over time as frequency synchronization emerges; (iv) genre clusters not coinciding with biological species clusters.

Expected for ritual cycle: (i) bimodal distribution of agent populations into those who have completed at least one ritual cycle and those who have not, with the former showing measurably different behavioral patterns through the AMEN-locked attributes; (ii) DivineNameAnchor accumulation in environment producing local resonance hot spots that other agents preferentially visit; (iii) similar success-rate statistics across the three registered ritual templates (Kabbalistic, Hesychast, Hindu), supporting the cross-cultural-structural-equivalence claim.

6.4 Limitations

The hands-off methodology is at odds with much of contemporary multi-agent research, which is heavily optimization-driven. This makes direct benchmark comparison impossible. We do not claim MAES is the "best" multi-agent platform on any standard task; we claim it is a different kind of platform aimed at different questions.

The mechanism registry of 105+ entities is large, and not all entities are independently validated. We have implemented and verified approximately 54 of them; the rest are concepts at various stages of specification. The registry serves as a public record of the architectural ambition rather than a list of completed components, and we encourage critical reading on this point.

The cross-cultural ritual claim deserves careful scrutiny. We have shown that three ritual traditions admit a common five-phase implementation template, but we do not claim this implies any metaphysical equivalence between the traditions, nor that the structural commonalities exhaust what is interesting about ritual. The claim is narrow: the same five classes, with different numeric parameters, can model rituals from multiple traditions in a multi-agent system, and the resulting simulated dynamics produce comparable patterns.

6.5 Reproducibility

All code, raw run data, and analysis scripts are released at `https://github.com/MAES-Lab/MAES`. The full Phase 1 extended protocol (90 runs total) takes approximately 12 hours on the hardware described in Section 4. A shorter pilot configuration (10 runs, ~95 minutes) is available for verification.

7 Conclusion and future work

MAES is offered as a research platform rather than a benchmark. Phase 1 results provide three findings that we view as the empirical foundation for the rest of the architectural program: (1) dimensional abduction is a large and reliable architectural lever, approximately doubling peak cognitive activity and producing a ninefold increase in species diversity; (2) explicit planning, at least in the form of fixed-horizon A over a learned world model, contributes less than 5% to ai_peak in either spatial regime and does not transfer between regimes; (3) the equilibrium distribution of cognitive styles is structurally determined by the search space, supporting the Cognitive Style Ecology Hypothesis.

The communication and ritual layers introduced in v3.2 open several research directions that we believe are novel: the integration of language and music as a single carrier with two channels, the emergence of cultural memory classes (songs, ballads, sagas, genres) independently of biological speciation, and the introduction of an ontologically-irreversible state-change operator into a multi-agent system. Empirical investigation of these layers is the subject of Phase 2 work, planned for the second half of 2026.

We invite collaboration, critique, and extension. The full mechanism registry, including the 30+ mechanisms not yet implemented, is included in the repository as an explicit invitation to other researchers who might want to take up specific entries.

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Note on bibliography: The reference list above is deliberately limited to works that we cite by name in the text. An extended bibliography covering related work in novelty search, quality-diversity, emergent communication, agent-based models of religion, and recent neural ALife will accompany Phase 2 publications, where the corresponding methods will be examined in more depth.